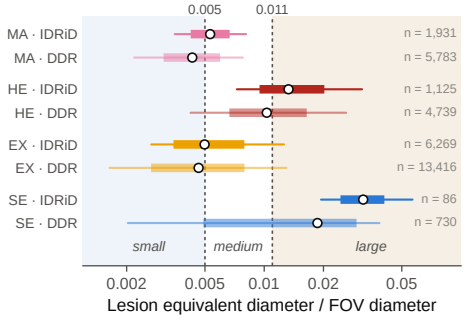
**a** **Lesion size by class**




Training images only (IDRiD 44, DDR 383). Bar, quartiles; line, 10th–90th percentile; dot, median.

**b** **Lesions that vanish when the image is shrunk**

**c** **Lesion-level recall by size**

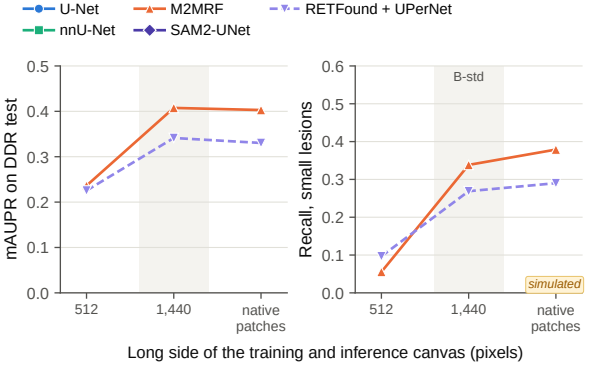
**Trained on IDRiD** — recall by lesion size

| | small | med. | large | prec. |
|---|---|---|---|---|
| HRNet-W48 | 0.58 | 0.88 | 0.86 | 0.48 |
| nnU-Net | 0.69 | 0.83 | 0.74 | 0.57 |
| HRDecoder | 0.54 | 0.80 | 0.92 | 0.66 |
| nnU-Net ResEnc-M | 0.60 | 0.85 | 0.70 | 0.53 |
| SAM2-UNet | 0.49 | 0.75 | 0.90 | 0.63 |
| WSRFNet | 0.49 | 0.76 | 0.90 | 0.63 |
| Swin-Unet | 0.67 | 0.85 | 0.83 | 0.42 |
| RETFound + UPerNet | 0.43 | 0.70 | 0.86 | 0.59 |
| DeepLabv3 | 0.27 | 0.53 | 0.78 | 0.55 |
| M2MRF | 0.62 | 0.82 | 0.74 | 0.53 |
| CLC-Net | 0.43 | 0.70 | 0.87 | 0.57 |
| MedSAM (full FT) | 0.39 | 0.66 | 0.84 | 0.57 |
| SAMed | 0.39 | 0.66 | 0.84 | 0.56 |
| U-Net | 0.65 | 0.85 | 0.80 | 0.49 |
| PMCNet | 0.38 | 0.66 | 0.85 | 0.56 |
| HACDR-Net | 0.41 | 0.75 | 0.76 | 0.50 |
| TC-Net | 0.42 | 0.69 | 0.86 | 0.58 |
| SAM-Adapter | 0.37 | 0.65 | 0.82 | 0.56 |
| FCT | 0.20 | 0.52 | 0.73 | 0.27 |

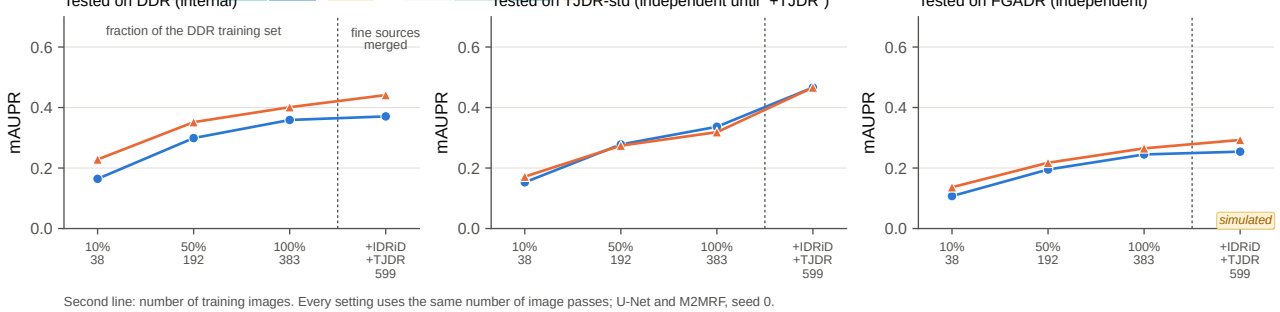
**Trained on DDR** — recall by lesion size

| | small | med. | large | prec. |
|---|---|---|---|---|
| HRNet-W48 | 0.22 | 0.49 | 0.68 | 0.51 |
| nnU-Net | 0.26 | 0.47 | 0.62 | 0.52 |
| HRDecoder | 0.27 | 0.56 | 0.78 | 0.50 |
| nnU-Net ResEnc-M | 0.29 | 0.49 | 0.66 | 0.49 |
| SAM2-UNet | 0.27 | 0.53 | 0.76 | 0.47 |
| WSRFNet | 0.26 | 0.54 | 0.76 | 0.47 |
| Swin-Unet | 0.31 | 0.54 | 0.67 | 0.42 |
| RETFound + UPerNet | 0.21 | 0.47 | 0.71 | 0.43 |
| DeepLabv3 | 0.06 | 0.29 | 0.66 | 0.50 |
| M2MRF | 0.28 | 0.56 | 0.70 | 0.45 |
| CLC-Net | 0.23 | 0.49 | 0.73 | 0.44 |
| MedSAM (full FT) | 0.19 | 0.44 | 0.67 | 0.45 |
| SAMed | 0.20 | 0.45 | 0.69 | 0.43 |
| U-Net | 0.25 | 0.48 | 0.65 | 0.44 |
| PMCNet | 0.21 | 0.47 | 0.71 | 0.44 |
| HACDR-Net | 0.11 | 0.35 | 0.56 | 0.42 |
| TC-Net | 0.19 | 0.42 | 0.67 | 0.41 |
| SAM-Adapter | 0.17 | 0.42 | 0.66 | 0.42 |
| FCT | 0.10 | 0.35 | 0.61 | 0.22 |

**d** **Input resolution**




**e** **Training amount and merged sources**




Second line: number of training images. Every setting uses the same number of image passes; U-Net and M2MRF, seed 0.